

ADAM ARZOUNI

Mechanical Engineer · Analytics & Automation · Data-Driven Operations

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PROFILE

Mechanical Engineer with 4+ years at Pratt & Whitney Canada, combining deep technical expertise with strong business impact. Built and deployed a production-grade Python analytics application that automated discrepancy claims processing, integrating OCR, SAP automation, and multi-user Excel reporting. Manages ~\$15M in monthly spare parts sales while designing internal data tools used daily by the team. Pursuing the McGill MMA to formalize expertise in machine learning and business intelligence — and pivot into a strategic analytics leadership role.

PROFESSIONAL EXPERIENCE

Customer Service Account Manager — Spare Parts | *Pratt & Whitney Canada*

Mar 2023 – Present

- ▶ RTX Workflow App (Python): Designed and deployed a production-grade desktop application automating discrepancy claims processing end-to-end — integrating multi-engine OCR (pytesseract + OpenCV), Outlook COM automation, SAP ZI40 parsing, SQLite database, and multi-user Excel logging with file-lock detection. Reduced processing time by ~80%.
- ▶ Built Power BI dashboards and Excel-based Gantt/KPI trackers to support operational planning using SAP data (MD62, BDI, CSR, BBR, AOG terminology).
- ▶ Managed ~\$15M in monthly spare parts sales as team lead for major Designated Overhaul Facilities (Lufthansa, Itochu, Covington) across Regional Airline and General Aviation markets.
- ▶ Hosted Quarterly Business Reviews (QBRs) evaluating ordering performance, delivery KPIs, and engine induction metrics for key global accounts.

Project Cost Analyst | *Pratt & Whitney Canada*

Jun 2022 – Mar 2023

- ▶ Consolidated 12+ distinct datasets into automated reporting dashboards, significantly reducing manual work and freeing team capacity for strategic analysis.
- ▶ Developed a Unigraph program to capture and analyze engine surge time and surge margin data; delivered actionable recommendations to engineering and supply chain stakeholders.
- ▶ Produced financial and operational reports for cross-functional teams spanning design, manufacturing, and cost analysis.

Manufacturing & Quality Engineering Analyst (Co-op) | *Pratt & Whitney Canada*

Jan 2020 – Apr 2020 | May 2021 – Aug 2021

- ▶ Designed automated data collection processes using SAP GUI API and VBA macros, generating ~\$20,000 in annual cost savings through measurable efficiency gains.
- ▶ Built Microsoft Access/VBA tools to streamline manufacturing data pipelines and monthly supplier performance reporting.
- ▶ Conducted root-cause investigations into dimensional variations on parts and measurement systems across multiple business units.

EDUCATION

B.Eng. — Mechanical Engineering | *École Polytechnique de Montréal, Université de Montréal* 2017 – 2022

Relevant coursework: Numerical Methods, Probability & Statistics, Systems Modeling & Simulation, Thermodynamics, Engineering Design

TECHNICAL SKILLS

Programming & Analytics: Python (pandas, OpenCV, pytesseract, pywin32, SQLite, CustomTkinter), SQL, VBA, C++, MATLAB/Simulink

Data & BI Tools: Power BI, Microsoft Excel (advanced), Microsoft Access, SAP (ZI40, MD62, BDI, AOG workflows), Unigraph, SCOAP

Engineering Software: CATIA V5, SolidWorks, AutoCAD, Nastran, Siemens NX, LabVIEW

Dev & Build Tools: PyInstaller, Git, openpyxl, PyMuPDF (fitz), COM automation, OCR pipeline design

Languages: French (native) · English (native) · Arabic (native) · Spanish (intermediate)

SELECTED PROJECTS

RTX Workflow App | *Python Desktop Analytics Tool — Pratt & Whitney Canada* 2022 – Present

- ▶ Production-grade CustomTkinter application for automated discrepancy claims management. Stack: multi-engine OCR with voting, Outlook COM email import + auto-reply, SAP ZI40 screenshot automation, SQLite, openpyxl multi-user Excel logging, Power BI Win32 reparenting.
- ▶ Engineered resilient PyInstaller build (COM threading isolation, network drive hang prevention, DPI fixes) — deployed daily by the Spare Parts Services team.

Phoenix Drone | *Engineering Competition — 1st Prize Winner* 2021

- ▶ Designed and built a rescue drone from scratch: custom electric motor, structural drawings, and full flight system integration. Awarded 1st Prize in inter-university competition.

Autonomous Robot Energy Connector | *Hydro-Québec / IREQ* 2020

- ▶ Designed a self-aligning mechanical connector system for autonomous robot recharging; simulated concept and assessed operating limits using SolidWorks and finite element analysis.
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ACTIVITIES & INTERESTS

Lasallian Movement for Environmental Protection (MLPE) · Scout Clan Member · Machine Automation & Robotics · Music